GEOMETRY · UNIT 3: TRIANGLE CONGRUENCE

Bridge Truss Inspector

Prove triangles congruent to inspect and design a footbridge truss.

CP2

2–3 class periods

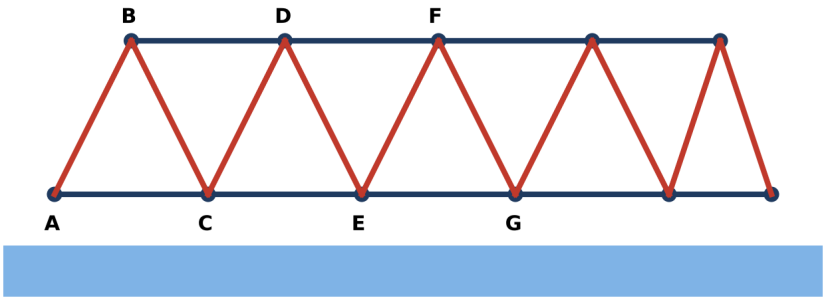
Name: _____ Date: _____ Period: _____

The Scenario

The town's new footbridge uses a Warren truss: a chain of triangles between a top beam and a bottom beam. Triangles are rigid, so a truss made of congruent triangles spreads the load evenly.

The inspector needs proof that the triangles are congruent, plus the missing measurements. Then the town wants your design for a second bridge.

Riverside Footbridge: Warren Truss



Learning Goals

- I can decide which congruence shortcut proves two triangles congruent.
- I can use corresponding parts to find unknown measures.
- I can complete a two-column proof.

Project Timeline

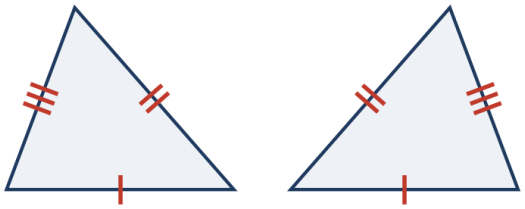
When	What you will do
Day 1	Inspect the Triangles
Day 2	Measure the Parts
Day 3	Design a Truss
Final	Truss Inspection Report + reflection

Part 1: Inspect the Triangles

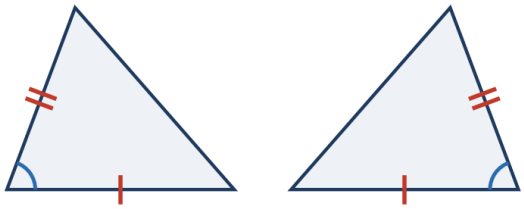
Day 1 · Look at the markings on each pair.

1. For each pair, name the shortcut (SSS, SAS, ASA, AAS) or write "not enough information".

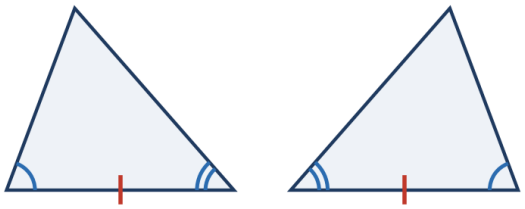
Pair 1



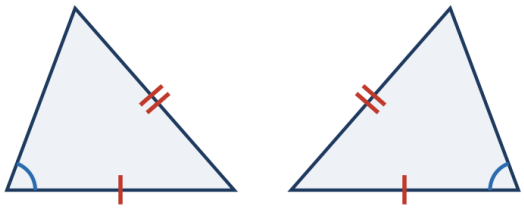
Pair 2



Pair 3



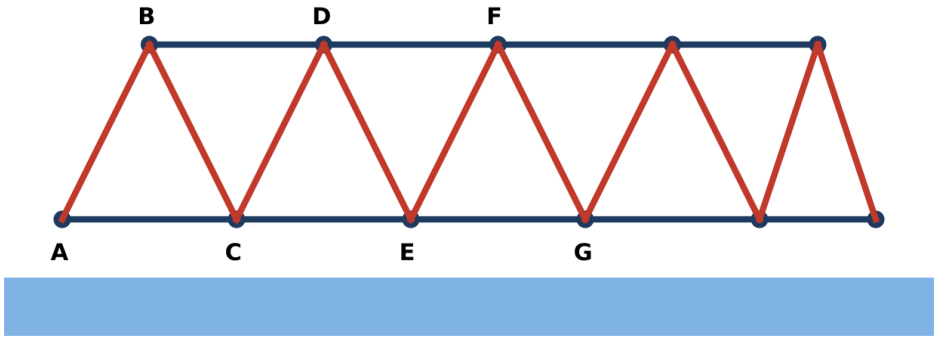
Pair 4



2. Why can't three pairs of congruent angles (AAA) prove two triangles congruent? Draw an example.

3. In the truss, AB ≅ CD, BC ≅ DE, and AC ≅ CE. Which shortcut proves △ABC ≅ △CDE?

Riverside Footbridge: Warren Truss



4. Explain why a triangle frame is rigid but a square frame can wobble.

Part 2: Measure the Parts

Day 2 · △ABC ≅ △DEF. Corresponding parts are congruent.

5. AB = (3x − 2) ft and DE = (2x + 5) ft. Find x and AB.

6. m∠B = (4y + 6)° and m∠E = 62°. Find y.

7. A truss triangle is isosceles with a vertex angle of 40°. Find each base angle.

8. Complete the proof. Given: AB ≅ CB and BD bisects ∠ABC. Prove: △ABD ≅ △CBD.

Statement	Reason
1. AB ≅ CB	Given
2. BD bisects ∠ABC	
3. ∠ABD ≅ ∠CBD	
4. BD ≅ BD	
5. △ABD ≅ △CBD	

Part 3: Design a Truss

Day 3 · Design a truss for a 40-foot bridge.

9. Sketch a truss with at least four triangles. Mark the congruent sides and angles.

10. Name the shortcut that proves your triangles are congruent.

11. If every triangle is equilateral with 10-foot sides, how many triangles touch the bottom beam? What is each angle?

Final Product: Truss Inspection Report

Your final product must include:

- Labeled truss diagram with congruence markings
- Shortcut decisions for each pair
- Solved measurements
- Completed proof
- Recommendation to the town

Reflection

R1. Why is SSS the reason triangles are used in bridges?

R2. Why must you name the vertices in matching order?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Truss design uses congruent triangles with correct markings and a valid congruence justification. Goes beyond the minimum with a creative, realistic choice.	Truss design uses congruent triangles with correct markings and a valid congruence justification.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Inspection report includes congruence decisions with shortcuts, solved measurements, at least one proof, and a labeled truss design. Polished, organized, and visually engaging.	Inspection report includes congruence decisions with shortcuts, solved measurements, at least one proof, and a labeled truss design. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.